

PRACTICAL COURSE WORKBOOK

SQL for Data Analysis

Write SQL reports whose rows and totals you can explain

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a customer order report with fixtures, expected results and reconciliation
SKILLS	SQL, PostgreSQL, Analytics, Joins, Critical evaluation, Documentation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Select and filter rows with explicit columns and ordering.
- Handle NULL values without confusing missing information with zero.
- Join and aggregate at the intended row grain.
- Use CTEs and window functions to explain an analytical query.
- Reconcile a final report against its input records.

WHO THIS IS FOR

Learners comfortable with spreadsheet rows and columns who want to write and check their own SQL reports.

BEFORE YOU BEGIN

- Understand rows, columns and basic arithmetic
- Access a PostgreSQL scratch database; most examples also run in SQLite

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Use SQL to answer practical questions about fictional orders. Filter records, handle missing values, join tables, group results and use window functions, then produce a report whose totals you can reconcile.

MODULE	LESSONS AND FOCUS
01 Read and filter records	1. Select columns and control row order 2. Filter records with explicit conditions 3. Treat missing amounts as unknown
02 Join and summarise	1. Count records and summarise known amounts 2. Keep customers without orders in a SQL report 3. Separate row filters from group filters
03 Explain analytical queries	1. Classify a record with CASE 2. Break an analysis into named query stages 3. Compare each sale with its customer total
04 Validate and deliver a report	1. Find duplicate identifiers before reporting 2. Reconcile included and excluded records 3. Deliver a report with zero-order customers included

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Read and filter records

Select columns and control row order; Filter records with explicit conditions; Treat missing amounts as unknown.

LESSON 01 / 18 MIN

Select columns and control row order

Learning objectives

- Select explicit columns.
- Identify what each result row represents.
- Use ORDER BY for a predictable result.

Key terms

Columns, Rows, Order

Retrieval prompt

What establishes a predictable order for this result?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Filter records with explicit conditions

Learning objectives

- Translate a request into row conditions.
- Distinguish AND from OR.
- Test a threshold and non-matching status.

Key terms

Predicate, Combination, Selection

Retrieval prompt

Which condition requires a sale to be both paid and at least 100?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Treat missing amounts as unknown

Learning objectives

- Find a missing value with IS NULL.
- Distinguish zero from unknown.
- Explain when a default is justified.

Key terms

NULL, Inspection, Policy

Retrieval prompt

Which predicate finds the fixture's missing amount?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Join and summarise

Count records and summarise known amounts; Keep customers without orders in a SQL report; Separate row filters from group filters.

LESSON 04 / 30 MIN

Count records and summarise known amounts

Learning objectives

- Distinguish row and value counts.
- Compute a known-value total.
- Report coverage beside an aggregate.

Key terms

Population, Coverage, Aggregate

Retrieval prompt

Why do COUNT(*) and COUNT(amount) differ here?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Keep customers without orders in a SQL report

Learning objectives

- Explain how an inner join differs from a left join.
- Count matching orders without counting an unmatched row.
- Check query output using a small, known dataset.

Key terms

exploration, SQL, data

Retrieval prompt

Which expression counts zero orders for an unmatched customer?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Separate row filters from group filters

Learning objectives

- Filter records before grouping.
- Compute one result per customer.
- Filter groups using an aggregate.

Key terms

WHERE, GROUP BY, HAVING

Retrieval prompt

Where does a condition on SUM(amount) belong in this grouped query?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

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3. REVIEW

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MODULE 03

Explain analytical queries

Classify a record with CASE; Break an analysis into named query stages; Compare each sale with its customer total.

LESSON 07 / 26 MIN

Classify a record with CASE

Learning objectives

- Write an ordered classification.
- Keep missingness distinct from a numeric band.
- Retain the original value beside a derived label.

Key terms

Condition, CASE, Label

Retrieval prompt

Why does the missing-value branch come first?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Break an analysis into named query stages

Learning objectives

- Name an intermediate query.
- Separate filtering from summarising.
- Inspect a stage without assuming materialisation.

Key terms

Fixture, CTE, Summary

Retrieval prompt

What does the paid CTE name in this example?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Compare each sale with its customer total

Learning objectives

- Keep individual sale rows in an analytical result.
- Define a window partition.
- Distinguish a window total from a grouped result.

Key terms

Row grain, Partition, Window total

Retrieval prompt

How many rows does this window query return for the four-row fixture?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

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3. REVIEW

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MODULE 04

Validate and deliver a report

Find duplicate identifiers before reporting; Reconcile included and excluded records; Deliver a report with zero-order customers included.

LESSON 10 / 22 MIN

Find duplicate identifiers before reporting

Learning objectives

- Check a candidate identifier for duplicates.
- Distinguish event keys from customer names.
- Avoid hiding duplicates with an unexplained DISTINCT.

Key terms

Identifier, Grouping, Duplicate check

Retrieval prompt

What does the repeated identifier check establish?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 11 / 26 MIN

Reconcile included and excluded records

Learning objectives

- Account for input records under an explicit rule.
- Count matching rows with a conditional expression.
- Check that included and excluded counts sum to the population.

Key terms

Population, Partitioned counts, Reconciliation

Retrieval prompt

With disjoint, exhaustive categories, what should their counts sum to?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

LESSON 12 / 30 MIN

Deliver a report with zero-order customers included

Learning objectives

- Preserve the complete customer population.
- Count orders and missing amounts separately.
- Explain and reconcile a final report.

Key terms

Population, Matching rules, Coverage report

Retrieval prompt

Why does Brian's displayed total of zero need the missing_amounts column?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a customer order report that includes customers with no orders.

DELIVERABLE	A runnable query, fixture records, expected-result table and reconciliation note.
TOOLS	PostgreSQL scratch database, SQL editor, Fictional customer and order fixtures

Production plan

- 1 Define the output grain as one row per customer.
- 2 Create a small fixture with multiple orders, one unmatched customer and a pending order.
- 3 Join and count paid orders while preserving customers with no matches.
- 4 Check missing values and whether filters change the intended population.
- 5 Reconcile counts and amounts to the source fixture and document the query and limitations.

Definition of done

- Every intended customer appears once.
- Unmatched customers have a zero matching-order count.
- Paid and pending orders are handled by an explicit rule.
- Report totals agree with an independent input calculation.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- What does one report row represent?
- Which rows were excluded and why?
- Can I explain every difference between source totals and report totals?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Add a fourth customer and an additional order, predict the new result and rerun the reconciliation.

PRIMARY REFERENCES

Sources to revisit

Querying a table

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/tutorial-select.html>

PostgreSQL SQL language guide

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/sql.html>

Value expressions and logical conditions

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/sql-expressions.html>

Comparison predicates

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/functions-comparison.html>

Aggregate functions

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/functions-aggregate.html>

Joins between tables

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/tutorial-join.html>

Aggregate functions in grouped queries

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/tutorial-agg.html>

Conditional expressions

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/functions-conditional.html>

WITH queries

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/queries-with.html>

Window functions

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/tutorial-window.html>

Grouped table expressions

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/queries-table-expressions.html>

SELECT processing and grouping

PostgreSQL Global Development Group - reviewed 2026-10-07

<https://www.postgresql.org/docs/current/sql-select.html>